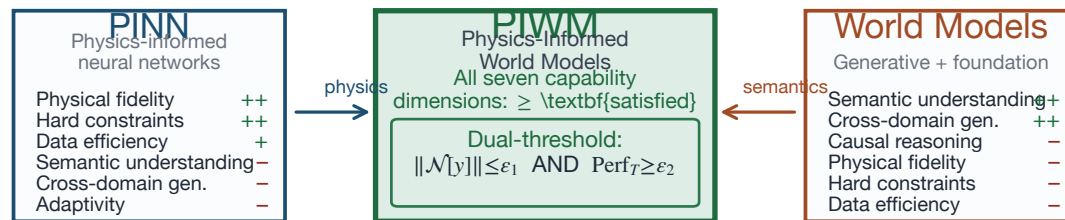


a The three-dimensional capability crisis



b Scissor argument: complementary ceilings unify in PIWM



c Four emergent capabilities resolve world-model ceilings 1:1



Fig. 1. (a) The three reinforcing failures that define the capability crisis. (b) PINN and world models occupy structurally complementary ceilings; PIWM unifies both and satisfies the dual-threshold condition. (c) Each emergent capability of PIWM resolves a specific world-model limitation.